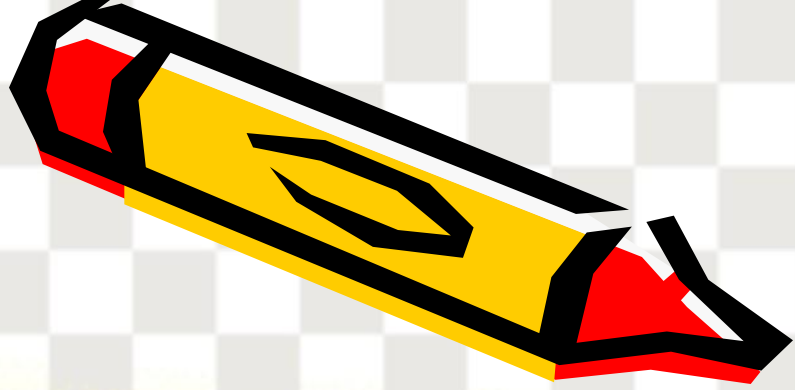
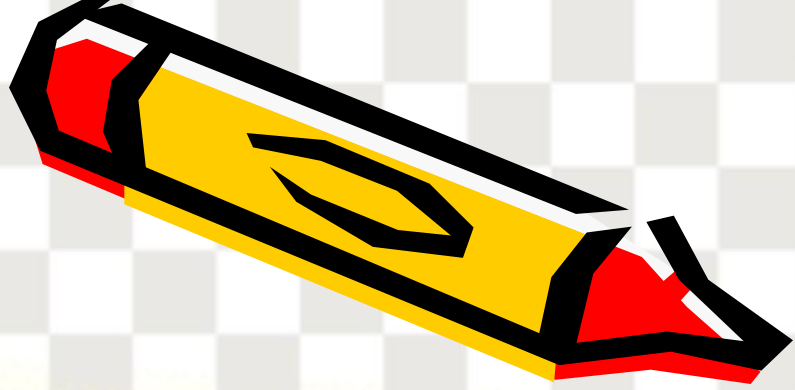




SUBSTANCE ABUSE DISORDERS



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EVIDENCE-BASED PRACTICE: Alcohol and Substance Use Disorders

The authors acknowledge that although effective treatments for alcohol and substance disorders do exist, they are not commonly and consistently practiced. This is particularly true for addition medications and psychosocial therapies. They would urge treatment programs to include these proven methods to enhance positive outcomes for clients and families.

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SUBSTANCE ABUSE DISORDERS

SUBSTANCE DEPENDENCE – A MALADAPTIVE PATTERN OF SUBSTANCE USE AS MANIFESTED BY 3 OR MORE OF THE FF:

1. **TOLERANCE** – increasing amount of the substance is required to achieve the desired effect or there is a markedly diminished effect with repeated use of the same dose.
2. **WITHDRAWAL** – the person following reduction or cessation of intake of the substance experiences a substance-specific syndrome. Such withdrawal signs could be physiologic or psychologic

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- ✗ When an individual persists in use of alcohol or other drugs despite problems related to use of the substance, substance dependence may be diagnosed

- - 3. The person continues to take the substance despite being aware that the substance causes physical & psychological problems
 - 4. A need for more of the substance than was intended
 - 5. The person is frequently under the influence of the substance or is intoxicated and is unable to stop using even when wanting to do so

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6. A great deal of time is spent in acquiring the substance or in recovering from its effects.
 7. Substance use causes social, occupational or recreational problems.

SUBSTANCE ABUSE

A. A maladaptive pattern of substance use leading to clinically significant impairment or distress as manifested by 1 or more of the ff:

- Failure to fulfill major role obligations at work, school or home
- Recurrent substance use in hazardous situations
- Recurrent substance related legal problems
- Continued substance use despite problems

B. Has never met the criteria for substance dependence for this class of substance

SUBSTANCE INTOXICATION

- A. The development of a substance specific syndrome due to a recent ingestion of a substance
- B. The signs & symptoms are not due to a general medical condition or is caused by another mental health disorder
- C. Clinically significant maladaptive behavioral or psychological changes due to the effect of the substance on the CNS

SUBSTANCE WITHDRAWAL

- A. The development of a substance specific signs & symptoms due to reduction or abstinence in the use of a substance
- B. The substance-specific syndrome causes clinically significant distress or impairment
- C. Not due to a general medical condition and not better accounted for by another mental disorder

THE *DSM-V-EDITION* LISTS DIAGNOSTIC CLASSES OF SUBSTANCE ABUSE:

- ✗ 1. Alcohol
- ✗ 2. Sedatives, hypnotics, and anxiolytics
- ✗ 3. Stimulants
- ✗ 4. Cannabis
- ✗ 5. Opioids
- ✗ 6. Hallucinogens
- ✗ 7. Inhalants

DETOXIFICATION

- ✗ Is the process of safely withdrawing from a substance.

ALCOHOLISM

WHO defines alcoholism as a “chronic disease or disorder or behavior characterized by alcohol consumption that exceeds customary use & interferes with the drinker’s health, interpersonal relations, or economic functioning

SYMPTOMS OF ALCOHOL DEPENDENCE

- ✗ Ingestion of larger amounts over a long period of time
- ✗ Inability to control or cut down drinking
- ✗ Frequent intoxication
- ✗ Withdrawal from social, occupational & recreational activities
- ✗ Marked tolerance
- ✗ Withdrawal symptoms
- ✗ Persistence of symptoms for at least one month

SIGNS OF ALCOHOLISM

SYSTEM INVOLVED	SIGNS & SYMPTOMS
INTEGUMENTARY	Spider angiomas, "dirty tan", flushed ruddy skin, mahogany finger stains, jaundice, acne rosacea, multiple bruises
MUSCULOSKELETAL	Unexplained fractures, muscle wasting of proximal muscle groups of lower & upper extremity
GENITOURINARY	Mild proteinuria, sexual dysfunctions, prostatitis, fluid & electrolyte imbalance
CARDIOVASCULAR	Erratic hypertensive course, paroxysmal atrial fibrillation, atrial tachycardia
NEUROLOGIC	Tremors that worsen with movement, vertigo & nystagmus, memory lapses, insomnia, seizures, hallucinations, peripheral neuropathy
RESPIRATORY	Frequent upper respiratory infection
GASTROINTESTINAL	Tooth losses, jaundice, cirrhosis, peptic ulcer, anorexia, wt loss, palpable liver, esophagitis, gastritis

- ✗ EPISODIC EXCESSIVE DRINKING -
characterized by intoxication less than 4x a year
- ✗ HABITUAL EXCESSIVE DRINKING –
characterized by intoxication for at least 12x a
year or once a week under alcohol influence

STAGES OF ALCOHOL ADDICTION

1. **PREADDICTION/EXPERIMENTAL USE** – the client drinks alcohol for the 1st or 2nd time & begins to learn about its effects
2. **RESPONSIBLE USE** – the client drinks alcohol in moderation; use does not harm the client or others

STAGES OF ALCOHOL ADDICTION

3. OCCASIONAL MISUSE ADDICTION – intake occasionally puts the client over the legal blood alcohol limit

STAGES OF ALCOHOL ADDICTION

4. EARLY PHASE (REGULAR MISUSE)

- ✗ Intake usually puts the client over the legal blood alcohol limit
- ✗ Drinking relieves such symptoms as anxiety
- ✗ Drinking helps the client cope
- ✗ The client is preoccupied with drinking
- ✗ Intake is rapid
- ✗ The client has hangovers
- ✗ The client minimizes intake when discussing it with others
- ✗ The client sneaks drinks
- ✗ Blackouts begin
- ✗ Personality changes occur

5. MIDDLE PHASE (DEPENDENCY)

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- × The client sets out to drink
- × Problems of alcohol abuse begin
- × The client begins to lie about drinking
- × The client begins to feel guilty about drinking
- × The client loses friends & interests
- × Blackouts increase
- × The client makes excuses for drinking
- × The client's alcohol tolerance increases
- × The client begins to lose control and can no longer predict what will happen once drinking starts

STAGES OF ALCOHOL ADDICTION

6. LATE PHASE (EXTREME OR COMPLETE DEPENDENCY)

- The client can not be without alcohol
- Physiologic symptoms develop if the client stops drinking
- The client drinks until supply is gone, then searches for more
- Alcohol becomes more important for the client than anything else
- The client drinks to feel "normal"

STAGES OF ALCOHOL ADDICTION

6. LATE PHASE (EXTREME OR COMPLETE DEPENDENCY)

- The client changes drinking places to hide the problem's severity from others
- Moral deterioration occurs
- Tolerance for alcohol decreases
- Thinking is impaired
- The client drinks in the morning
- The client may change jobs or move to another city to "start over again" only to resume the drinking pattern (geographic cure)

LEVELS OF ALCOHOL INTOXICATION

- × **Mild Intoxication**
- × (BAL of 0.1% - 0.2%)
- ×
- × Slurred speech and talkativeness
- × Loss inhibition
- × Unsteady gait
- × Motor incoordination

LEVELS OF ALCOHOL INTOXICATION

- ✗ Moderate Intoxication
- ✗ (BAL of 0.2% - 0.3%)
- ✗ Blackouts
- ✗ Ataxia
- ✗ Impaired memory (confabulation)
- ✗ Tremors

LEVELS OF ALCOHOL INTOXICATION

- ✗ **Severe Intoxication**
- ✗ (BAL of 0.3% and above)
- ✗
- ✗ Coma
- ✗ Altered level of consciousness
- ✗ Respiratory depression
- ✗ Stupor

WITHDRAWAL SYMPTOMS

1. **ALCOHOLIC HALLUCINOSIS** - occurs after a sudden cessation of alcohol intake. The patient experiences auditory hallucinations, illusions & vivid frightening dreams that are terrifying. The patient appears similar to someone who is in the acute stage of schizophrenia. However, consciousness remains clear, & the signs of autonomic lability seen in delirium tremens are usually absent. Often hallucinations precedes delirium tremens. Recovery takes 1-3 wks and recurrence is common if patient resumes drinking.

WITHDRAWAL SYMPTOMS

2. **DELIRIUM TREMENS** - occurs 48-72 hrs after the last drink. Patient experiences massive anxiety, confusion, insomnia & nightmares if able to sleep, diaphoresis, elevated temperature, tachycardia. Profound depression, hallucinations, fear & even terror also occur. Often during the initial stage, the person has a false belief that he is back to a habitual activity such as his work. It is during delirium tremens that patients complain that the floor is moving, the walls are falling, the room is rotating, insects are crawling in his body & resting tremors of the hand develops. This is self limiting & usually resolves within 12-24 hrs after onset, if no other disorder may be present

WITHDRAWAL SYMPTOMS

3. THIAMINE DEFICIENCY - an intoxicated patient is at risk for hypoglycemia. If a patient is suspected of being a chronic alcoholic, 100 mg of thiamine is administered along with the parenteral dextrose. This is because dextrose, if given alone, could deplete cerebral stores of thiamine & cause Wernicke's encephalopathy. Vitamin B1 is given parenterally because the amount needed for replenishment can not be absorbed enterally either in sufficient time or quantity to prevent Wernicke-Korsakoff's syndrome

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- ✗ To promote safe withdrawal include Benzodiazepine to suppress the withdrawal symptoms
 - ✗ Valium (Diazepam)
 - ✗ Ativan (Lorazepam)
 - ✗ Librium (Chlordiazepoxide)
 - ✗

COMPLICATIONS

1. WERNICKE'S PSYCHOSIS/ENCEPHALOPATHY - an inflammatory, hemorrhagic, degenerative condition of the brain caused by thiamine deficiency and Vit B12. Symptoms include double vision, involuntary & rapid eye movements, lack of muscular coordination & decreased mental function, may be mild or severe. Intervention is to give Thiamine 100 mg IM followed by 50 mg po daily, plus B12 & folate both 1 mg/day po, should be given to prevent Wernicke-Korsakoff syndrome
2. KORSAKOFF'S PSYCHOSIS - is sometimes preceded by Wernicke's encephalopathy. Confusion & ataxia are predominant symptoms. Other symptoms are short term memory amnesia, inability to learn new skills, disorientation & confabulation. Irreversible. Large doses of thiamine may prevent the development of this.

COMMON BEHAVIORAL PROBLEMS SEEN IN ALCOHOLICS

- ✗ Destructive and rebellious behavior
- ✗ Dominant and critical behavior
- ✗ Difficulty with intimate relationships and tendency toward narcissism
- ✗ Decreased self-esteem

TREATMENT FOR ALCOHOLISM

- × **FOLIC ACID (Folate)**

- + Treat nutritional deficiency

- × **THIAMINE (Vit. B1)**

- + Prevent or treat Korsakoff-Wernicke's Syndrome

- × **CYANOCOBALAMIN (Vit. B12)**

- + Prevent or treat nutritional deficiencies

- × **BENZODIAZEPINE**

- + Suppress the symptoms of abstinence

TREATMENT FOR ALCOHOLISM

✖ NALTREXONE (Revia, Trexan)

- + Used to reduce alcohol craving
- + Used for the treatment of opioid abuse (blocks the effects of opiates)

✖ DISULFIRAM (Antabuse)

- + Helps client to maintain abstinence from alcohol

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- ✗ ALCOHOLICS ANONYMOUS

- ✗

- ✗ An international mutual aid movement claiming more than 2 million members and declaring its “primary purpose is to stay sober and help other alcoholics achieve sobriety”

NARCOTIC/OPIOID ADDICTION

Narcotics or opioids are a group of drugs generally used as pain relievers. The most commonly abused opioids are:

- Heroin - illegal & most addictive drug
- Morphine - a major pain reliever
- Codeine - found in cough suppressants
- Demerol - a major pain reliever
- Paregoric - pain reliever
- Opium

NARCOTIC/OPIOID ADDICTION

- Also called "white stuff", "hard stuff", junk
- Administered by **skin injection** (skin popping), IV (main line), inhalation (snorted), orally. Tell tale signs are needle tracks or inflamed nasal mucosa if snorted.
- Emotional dependence occurs 1st followed by physical dependence.
- Physical dependence is characterized by the need to have 3 or more narcotic injections a day, the presence of fresh needle marks, presence of withdrawal symptoms & signs & the presence of morphine glucoronide in urine specimen.
- Heroin is biotransformed to morphine, conjugated with glucoronide & excreted.

NARCOTIC/OPIOID ADDICTION

- Opioids induce cross-tolerance so that abusers can substitute one for another. When tolerance develops, the person must take a higher dosage to achieve the desired effect.
- Since most opioids are pain relievers, persons who abuse this substance usually require higher dosage of pain relievers such as anesthesia during surgery

NARCOTIC/OPIOID - WITHDRAWAL

- ✗ Symptoms that develop as withdrawal progresses
- ✗ include nausea, vomiting, dysphoria, lacrimation,
- ✗ rhinorrhea, sweating, diarrhea, yawning, fever, and
- ✗ insomnia. Symptoms of opioid withdrawal cause significant
- ✗ distress but do not require pharmacologic intervention
- ✗ to support life or bodily functions

HALLUCINOGENS/PSYCHO-MIMETICS/PSYCHEDELICS

Hallucinogens cause CNS excitation & central autonomic hyperactivity that results in alteration in perception (hallucinations) and mood (usually euphoric, sometimes depressive)

2 BASIC GROUPS:

I. NATURAL - blocks reuptake of serotonin

1. Mescaline- takes effect within 30 minutes & lasts for 12 hrs. Addicts taking this substance report that colors become more vivid, music more beautiful & sounds more intense, they become insightful; good trip is characterized as hilarity & joy; bad trip is panic-paranoid thinking & anxiety

HALLUCINOGENS

- ✗ mescaline, psilocybin, lysergic acid diethylamide
- ✗ (LSD), and “designer drugs” such as Ecstasy.
- ✗ Phencyclidine (PCP),

HALLUCINOGENS

2. **PSILOCYBIN** - onset is within 25 minutes & effect is up to 8 hrs. A sense of unreality occur characterized by hallucinations & illusions. Death may occur due to accidents caused by perceptual distortions. Dilated pupils, tachycardia, increased BP & temperature are observed
3. **MARIJUANA / "POT" or HASHISH** - effect lasts for 4 hrs when smoked & 12 hrs when taken orally. Stays in the blood for 4 wks after ingestion or inhalation. Chronic use results in psychologic dependence but causes no physical dependence. Induces euphoria characterized by a dreamy state of consciousness in which person experiences a keener sight & hearing, ideas seem disconnected yet free flowing & unanticipated, perception of time is altered, increased appetite, relief of nausea & vomiting, impaired balance, memory, concentration & decision making ability. Dry mouth, sore throat, tachycardia, dilated pupils & conjunctival irritation are also observed

HALLUCINOGENS

II. SYNTHETIC OR SEMI-SYNTHETIC

1. **LYSERGIC ACID DIETHYLAMIDE (LSD)** - take effect in 40 minutes & lasts for 12 hrs. Causes synesthesia (blending of senses), user reports of tasting a sound or smelling a color. Gives a sense of unreality, perceptual alterations & distortions, dilated pupils, increased BP, tachycardia, trembling & impaired judgment. Many have committed suicide because of the bad trip from LSD.
- Many LSD users experience flashbacks long after they have terminated use of this drug. The **flashbacks consist of visual illusions & hallucinations**. Often the flashback is precipitated by the use of alcohol, marijuana, barbiturates or by stress and fatigue. Sometimes it just occurs out of the blue with no apparent reason. Flashbacks tend to subside within 6-12 months

HALLUCINOGENS

II. SYNTHETIC OR SEMI-SYNTHETIC

2. PHENCYCLIDINE (PCP) - first known as the PEACE PILL because it was initially used as anesthetic for humans. PCP can be administered by injection or smoking (frequently sprayed on a smoking material such as tobacco, mint, parsley or marijuana).

Patients brought to the ER under this drug exhibits unpredictable manifestations that of alternating violence and coma. Good trip include euphoria, & a peaceful easy feeling. Perceptual distortions are common. Elevated vital signs, ataxia, salivation, vomiting, catatonic muscular rigidity, alternating with violent outbursts, blank stare & agitation are observed.

CNS STIMULANTS

Blocks reuptake of norepinephrine (Amphetamine) and dopamine (cocaine) that results in prolonged effect of this neurotransmitters resulting in euphoric excitement. CNS stimulants include:

- Amphetamines
- Benzedrine inhalers
- Caffeine
- Cocaine (crack)
- Diet pills
- Ecstasy (Methylenedioxymethamphetamine)

CNS DEPRESSANTS

CNS depressants, which are also referred to as anxiolytic or hypnotic drugs include:

- Barbiturates
- Benzodiazepines
- Methaqualone (Quaaludes, "snoopers")

These drugs produce both psycho logic & physical dependence. Tolerance is common and cross-tolerance with ethanol (found in alcoholic drinks) occurs. Most people who abuse these drugs prefer the rapid onset drugs such as the secobarbital or pentobarbital. These drugs are known to cause the ff effects:

- depressed reflexes, sense of calmness, slightly diminished alertness, lack of coordination & postural unsteadiness, slurred speech, decreased inhibitions, nystagmus, small pupils, respiratory depression, poor memory & impaired judgment, slow thinking & narrowed attention span, emotional lability

INHALANTS

- ✗ **Inhalants** are a diverse group of drugs including
- ✗ anesthetics, nitrates, and organic solvents

EXAMPLE: aromatic hydrocarbons found in gasoline, glue, paint thinner, and spray paint.

INTERVENTION

✕ PROVIDING HEALTH TEACHING FOR CLIENT
AND FAMILY

ADDRESSING FAMILY ISSUES

✕ PROMOTING COPING SKILLSPROMOTING COPING SKILLS

SUBSTANCE ABUSE IN HEALTH PROFESSIONALS

- ✗ Incorrect drug counts
- ✗ • Excessive controlled substances listed as wasted or contaminated
- ✗ • Reports by clients of ineffective pain relief
- ✗ from medications especially if relief had been adequate previously
- ✗ • Damaged or torn packaging on controlled substances
- ✗ • Increased reports of “pharmacy error”

SUBSTANCE ABUSE IN HEALTH PROFESSIONALS

- ✗ • Consistent offers to obtain controlled substances from pharmacy
- ✗ • Unexplained absences from the unit
- ✗ • Trips to the bathroom after contact with controlled substances
- ✗ • Consistent early arrivals at or late departures from work for no apparent reason

✕ Thank you